Exercise Sheet

Lecture 4

Matrices of Linear Maps; Change of Basis

Throughout, bases are ordered and $\mathcal{M}(T)$ denotes the matrix of T with respect to the bases in use. Tags indicate the flavour of each problem; a $\star$ marks a harder one.

Core tutorial route

Work **4.1, 4.7, 4.11, 4.14, 4.15, 4.16, 4.17, and 4.23** in that order. This route covers building the matrix of a map, commuting operators and their matrices, testing invertibility, the dimension of a map space, the equivalence of injectivity, surjectivity and invertibility, computing a matrix in a new basis, converting coordinates, and the proof that trace and determinant are invariant.

Additional practice: 4.3–4.4, 4.8–4.10, 4.12–4.13, 4.18–4.22. **Bridge:** 4.2, 4.5, 4.6 (differentiating polynomials). **Extension:** 4.24–4.26.

Matrices of maps

Exercise 4.1 (computational) Let $T \in \mathcal{L}(\mathbb{R}^3, \mathbb{R}^2)$ be given by $T(x, y, z) = (x + 2y - z,\ 3x + z)$. Write down the matrix $\mathcal{M}(T)$ with respect to the standard bases and use it to compute $[T(1, 1, 1)]$.

Answer. $\mathcal{M}(T) = \left(\begin{smallmatrix} 1 & 2 & -1 \\ 3 & 0 & 1 \end{smallmatrix}\right)$; $[T(1, 1, 1)] = (2, 4)$.

Exercise 4.2 (computational) Let $D \in \mathcal{L}(\mathcal{P}_2(\mathbb{R}))$ be the differentiation operator. In the standard basis $1, t, t^2$ compute the matrices $\mathcal{M}(D)$ and $\mathcal{M}(D)^2$. Which operator has the matrix $\mathcal{M}(D)^2$?

Answer. $\mathcal{M}(D) = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{smallmatrix}\right)$, $\mathcal{M}(D)^2 = \left(\begin{smallmatrix} 0 & 0 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{smallmatrix}\right) = \mathcal{M}(D^2)$.

Exercise 4.3 (computational) Let $T \in \mathcal{L}(\mathbb{R}^2)$ be given by $T(x, y) = (2x - y,\ x + 3y)$. Write down the matrix $\mathcal{M}(T)$ in the standard basis and use it to compute $[T(1, 2)]$.

Answer. $\mathcal{M}(T) = \left(\begin{smallmatrix} 2 & -1 \\ 1 & 3 \end{smallmatrix}\right)$; $[T(1, 2)] = (0, 7)$.

Exercise 4.4 (computational) Let $T \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^3)$ be given by $T(x, y) = (x - y,\ 2x + y,\ 4y)$. Write down the matrix $\mathcal{M}(T)$ with respect to the standard bases and use it to compute $[T(3, -1)]$.

Answer. $\mathcal{M}(T) = \left(\begin{smallmatrix} 1 & -1 \\ 2 & 1 \\ 0 & 4 \end{smallmatrix}\right)$; $[T(3, -1)] = (4, 5, -4)$.

Exercise 4.5 (computational) Let $T \in \mathcal{L}(\mathcal{P}_2(\mathbb{R}))$ be given by $(Tp)(t) = p(t) + t\,p'(t)$. In the standard basis $1, t, t^2$ write down the matrix $\mathcal{M}(T)$ and use it to compute $T(2 - t + 3t^2)$.

Answer. $\mathcal{M}(T) = \operatorname{diag}(1, 2, 3)$; $T(2 - t + 3t^2) = 2 - 2t + 9t^2$.

Exercise 4.6 (computational) Let $D \in \mathcal{L}(\mathcal{P}_3(\mathbb{R}))$ be the differentiation operator. Its matrix in the standard basis $1, t, t^2, t^3$ is

$$\mathcal{M}(D) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Compute $\mathcal{M}(D)^3$ and find the smallest k such that $D^k = 0$.

Answer. $\mathcal{M}(D)^3 = \begin{pmatrix} 0&0&0&6\\ 0&0&0&0\\ 0&0&0&0\\ 0&0&0&0 \end{pmatrix}$; the smallest k is $k = 4$.

Exercise 4.7 (proof) Let $\dim V = n$, and let $S, T \in \mathcal{L}(V)$ be operators whose matrices in a fixed basis are $\mathcal{M}(S), \mathcal{M}(T)$. Prove that the operators commute, $ST = TS$, if and only if their matrices commute, $\mathcal{M}(S)\mathcal{M}(T) = \mathcal{M}(T)\mathcal{M}(S)$.

Hint. The theorem "A matrix acts on coordinates", applied twice, gives $\mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T)$. Then use the fact that the map $\mathcal{M}$ is injective: operators are equal if and only if their matrices are equal.

Exercise 4.8 (proof) Fix bases of V and W. Prove that for any $S, T \in \mathcal{L}(V, W)$ and $a \in \mathbb{F}$, $\mathcal{M}(aS + T) = a\,\mathcal{M}(S) + \mathcal{M}(T)$.

Hint. Compare the matrices column by column. The k-th column of $\mathcal{M}(aS + T)$ is the coordinate vector of $(aS + T)v_k$, and the coordinate map is linear.

Invertibility and isomorphism

Exercise 4.9 (computational) Decide whether the operator $T(x, y) = (2x + y,\ 4x + 2y)$ on $\mathbb{R}^2$ is invertible. If it is, find T^{-1}. If not, find a nonzero vector in the kernel $\ker T$.

Answer. Not invertible ($\det = 0$); $(1, -2) \in \ker T$.

Exercise 4.10 (computational) Show that the polynomials $1,\ 1 - t,\ (1 - t)^2$ form a basis of $\mathcal{P}_2(\mathbb{R})$. Transport the question to $\mathbb{R}^3$.

Answer. The coordinate vectors $(1, 0, 0)$, $(1, -1, 0)$, $(1, -2, 1)$ are linearly independent ($\det = -1$), so the polynomials form a basis.

Exercise 4.11 (computational) Decide whether the operator $T(x, y) = (3x + 5y,\ x + 2y)$ on $\mathbb{R}^2$ is invertible. If it is, write down a formula for T^{-1}.

Answer. Invertible ($\det = 1$); $T^{-1}(x, y) = (2x - 5y,\ -x + 3y)$.

Exercise 4.12 (computational) Decide whether the operator $T(x, y, z) = (x + 2y + 3z,\ 2x + 4y + 6z,\ x + y + z)$ on $\mathbb{R}^3$ is invertible. If not, find a nonzero vector in the kernel $\ker T$.

Answer. Not invertible ($\det = 0$); $(1, -2, 1) \in \ker T$.

Exercise 4.13 (computational) Show that the polynomials $1 + t^2,\ t + t^3,\ 1 + t,\ t^2$ form a basis of $\mathcal{P}_3(\mathbb{R})$. Transport the question to $\mathbb{R}^4$.

Answer. The determinant of the coordinate vectors is $1 \neq 0$, so the polynomials are linearly independent and form a basis.

Exercise 4.14 (proof) Prove that if V and W are finite-dimensional, then $\dim \mathcal{L}(V, W) = (\dim V)(\dim W)$.

Hint. Show that the map $\mathcal{M}\colon \mathcal{L}(V, W) \to \mathbb{F}^{m \times n}$ is an isomorphism (a matrix prescribes exactly the images of the basis vectors), and compute the dimension of the matrix space.

Exercise 4.15 (proof) Let V be finite-dimensional and let $T \in \mathcal{L}(V)$. Prove that the following are equivalent: (i) T is injective; (ii) T is surjective; (iii) T is invertible.

Hint. Use rank–nullity: $\dim \ker T + \dim \operatorname{im} T = \dim V$.

Change of basis and similarity

Exercise 4.16 (computational) Let $T(x, y) = (x, -y)$ be the reflection in the x-axis on $\mathbb{R}^2$. Its matrix in the standard basis is $A = \operatorname{diag}(1, -1)$. Using the formula $A' = P^{-1}AP$, find the matrix of this reflection in the basis $\mathcal{B}' = \big((1, 1), (1, -1)\big)$.

Answer. $A' = P^{-1}AP = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$.

Exercise 4.17 (computational) Let $\mathcal{B}$ be the standard basis of $\mathbb{R}^2$ and $\mathcal{B}' = \big((2, 1), (1, 1)\big)$. The coordinate vector of a vector v in the new basis is $[v]_{\mathcal{B}'} = (1, 2)$. Find $[v]_{\mathcal{B}}$ (that is, the vector v itself). Also find the coordinate vector $[w]_{\mathcal{B}'}$ of the vector $w = (3, 0)$.

Answer. $[v]_{\mathcal{B}} = (4, 3)$; $[w]_{\mathcal{B}'} = (3, -3)$.

Exercise 4.18 (computational) The matrix of an operator $T \in \mathcal{L}(\mathbb{R}^2)$ in the standard basis is $A = \left(\begin{smallmatrix} 2 & 1 \\ 0 & 3 \end{smallmatrix}\right)$. Using the formula $A' = P^{-1}AP$, find its matrix in the basis $\mathcal{B}' = \big((1, 0), (1, 1)\big)$.

Answer. $A' = \operatorname{diag}(2, 3)$.

Exercise 4.19 (computational) In $\mathbb{R}^2$ take the old basis $\mathcal{B} = \big((1, 2), (1, 3)\big)$ and the new basis $\mathcal{B}' = \big((2, 5), (-1, -2)\big)$. A vector v has the coordinate vector $[v]_{\mathcal{B}} = (1, 3)$, and an operator $T \in \mathcal{L}(\mathbb{R}^2)$ has the matrix $\mathcal{M}(T, \mathcal{B}) = \left(\begin{smallmatrix} 0 & 1 \\ 2 & 1 \end{smallmatrix}\right)$. Find the change-of-basis matrix P, the coordinate vector $[v]_{\mathcal{B}'}$ and the matrix $\mathcal{M}(T, \mathcal{B}')$.

Answer. $P = \left(\begin{smallmatrix} 1 & -1 \\ 1 & 0 \end{smallmatrix}\right)$; $[v]_{\mathcal{B}'} = (3, 2)$; $\mathcal{M}(T, \mathcal{B}') = \left(\begin{smallmatrix} 3 & -2 \\ 2 & -2 \end{smallmatrix}\right)$.

Exercise 4.20 (computational) In $\mathbb{R}^3$ take the old basis $\mathcal{B} = \big((1, 0, 1), (0, 1, 1), (1, 1, 1)\big)$ and the new basis $\mathcal{B}' = \big((1, 1, 0), (0, 1, 1), (0, 0, 1)\big)$. Find the change-of-basis matrix P and the coordinate vector $[v]_{\mathcal{B}'}$ of the vector v with $[v]_{\mathcal{B}} = (2, -1, 1)$.

Answer. $P = \left(\begin{smallmatrix} -1 & 0 & 1 \\ -1 & 1 & 1 \\ 2 & 0 & -1 \end{smallmatrix}\right)$; $[v]_{\mathcal{B}'} = (3, -3, 5)$.

Exercise 4.21 (computational) In $\mathcal{P}_1(\mathbb{R})$ take the old basis $\mathcal{B} = (1 - t, 2 - t)$ and the new basis $\mathcal{B}' = (1 + t, t)$. An operator $T \in \mathcal{L}(\mathcal{P}_1(\mathbb{R}))$ has the matrix $\mathcal{M}(T, \mathcal{B}) = \left(\begin{smallmatrix} -1 & -1 \\ 2 & 3 \end{smallmatrix}\right)$. Find $\mathcal{M}(T, \mathcal{B}')$.

Answer. $\mathcal{M}(T, \mathcal{B}') = \left(\begin{smallmatrix} 1 & -1 \\ -2 & 1 \end{smallmatrix}\right)$.

Exercise 4.22 (computational) Using only trace and determinant, decide whether the matrices in each pair can be similar:

$$\text{(a)}\quad \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \text{ and } \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}; \qquad \text{(b)}\quad \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} \text{ and } \begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix}.$$

Answer. (a) No: the traces differ, $5 \neq 4$. (b) The invariants do not rule similarity out: both matrices have trace 4 and determinant 4.

Exercise 4.23 (proof) Prove that similar matrices have the same determinant and the same trace.

Hint. For the determinant, use $\det(XY) = \det X \det Y$. For the trace, use the identity $\operatorname{tr}(XY) = \operatorname{tr}(YX)$, grouping the product $P^{-1}AP$ as $(P^{-1}A)P$.

Exercise 4.24 ($\star$) Suppose the matrix A is similar to the matrix B, and B is similar to C. Prove that A is similar to C. Then prove that similarity is an equivalence relation on the set of $n \times n$ matrices.

Hint. For transitivity, multiply the two matrices that give the similarities and simplify the expression $(PQ)^{-1}A(PQ)$.

Exercise 4.25 ($\star$) Suppose A and B are $n \times n$ matrices and A is invertible. Prove that the matrices AB and BA are similar. Then give an example showing that without the invertibility assumption the matrices AB and BA need not be similar.

Hint. You need a matrix P such that $P^{-1}(AB)P = BA$. Since A is invertible, try $P = A$. For the example, look for 2×2 matrices whose product is the zero matrix in one order but not in the other.

Exercise 4.26 ($\star$) Give two 2×2 matrices whose traces and determinants agree, but which are *not* similar. Prove that they are not similar.

Hint. Compare the matrix $3I$ with a non-diagonal matrix that has the same trace and determinant. What is the product $P^{-1}(3I)P$?

Answer. For example, $\left(\begin{smallmatrix} 3 & 0 \\ 0 & 3 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 3 & 1 \\ 0 & 3 \end{smallmatrix}\right)$: both have trace 6 and determinant 9, but they are not similar.